WORKED SOLUTION

STEP I 1990, Question 8

Pure • Infinite Trigonometric Products

STEP I 1990, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q8

The key idea

The double-angle formula $\sin 2\theta = 2\sin\theta\cos\theta$ turns each cosine into a ratio of two sines, $\cos\theta = \dfrac{\sin 2\theta}{2\sin\theta}$. Multiply such ratios together and almost everything cancels, like a telescoping sum. Once the finite product is a single neat fraction, letting $n \to \infty$ only needs the standard limit $\dfrac{\sin x}{x} \to 1$ as $x \to 0$.

Solution

A word about the condition on α

Every step below divides by $\sin(\alpha/2^m)$ for some $m \geq 0$, so we need these never to vanish. If $\sin(\alpha/2^m) = 0$ then $\alpha/2^m = j\pi$ for some integer j, so $\alpha = (2^m j)\pi$ is an integer multiple of π, which is excluded. So $\sin(\alpha/2^m) \neq 0$ for all $m \geq 0$.

The product of two cosines

Applying $\sin 2\theta = 2\sin\theta\cos\theta$ twice, first with $\theta = \alpha/2$ and then with $\theta = \alpha/4$,

$$\begin{aligned}\sin\alpha &= 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}\\ &= 2\left(2\sin\frac{\alpha}{4}\cos\frac{\alpha}{4}\right)\cos\frac{\alpha}{2} = 4\sin\frac{\alpha}{4}\cos\frac{\alpha}{2}\cos\frac{\alpha}{4}.\end{aligned}$$

Since $\sin(\alpha/4) \neq 0$ we may divide by $4\sin(\alpha/4)$, which gives

$$\cos\frac{\alpha}{2}\cos\frac{\alpha}{4} = \frac{\sin\alpha}{4\sin(\alpha/4)},$$

as required.

The product of n cosines

Write $P_n = \cos\frac{\alpha}{2}\cos\frac{\alpha}{4}\cdots\cos\frac{\alpha}{2^n}$. We prove by induction on n that

$$P_n = \frac{\sin\alpha}{2^n\sin(\alpha/2^n)}.$$

For $n = 1$ this reads $\cos\frac{\alpha}{2} = \dfrac{\sin\alpha}{2\sin(\alpha/2)}$, which is the double-angle formula divided by $2\sin(\alpha/2) \neq 0$. (The case $n = 2$ is the result we have just shown.)

Now suppose the formula holds for some $n \geq 1$. The double-angle formula with $\theta = \alpha/2^{n+1}$ gives $\sin\frac{\alpha}{2^n} = 2\sin\frac{\alpha}{2^{n+1}}\cos\frac{\alpha}{2^{n+1}}$. Hence

$$\begin{aligned} P_{n+1} = P_n\cos\frac{\alpha}{2^{n+1}} &= \frac{\sin\alpha}{2^n\sin(\alpha/2^n)}\cos\frac{\alpha}{2^{n+1}} \\ &= \frac{\sin\alpha\,\cos(\alpha/2^{n+1})}{2^n\cdot 2\sin(\alpha/2^{n+1})\cos(\alpha/2^{n+1})} = \frac{\sin\alpha}{2^{n+1}\sin(\alpha/2^{n+1})}. \end{aligned}$$

Cancelling $\cos(\alpha/2^{n+1})$ is legitimate: it is a factor of $\sin(\alpha/2^n)$, which is non-zero, so it is non-zero too. This is the formula for $n+1$, so by induction it holds for every positive integer n.

Letting $n \to \infty$

The infinite product means the limit of the partial products P_n. Put $x_n = \alpha/2^n$, so $x_n \to 0$ as $n \to \infty$, and

$$2^n\sin\frac{\alpha}{2^n} = \alpha\cdot\frac{\sin x_n}{x_n} \longrightarrow \alpha\cdot 1 = \alpha.$$

? **Ask yourself** Why is it safe to turn $\lim P_n = \dfrac{\sin\alpha}{\alpha}$ upside down?

Because the limit is not zero: $\alpha \neq 0$ and $\sin\alpha \neq 0$ for our α. So $P_n \to \dfrac{\sin\alpha}{\alpha} \neq 0$, the infinite product exists, and dividing $\sin\alpha$ by it gives

$$\frac{\sin\alpha}{\cos\frac{\alpha}{2}\cos\frac{\alpha}{4}\cos\frac{\alpha}{8}\cdots} = \lim_{n\to\infty}\frac{\sin\alpha}{P_n} = \lim_{n\to\infty}2^n\sin\frac{\alpha}{2^n} = \alpha,$$

which is the required formula.

The formula for $\pi/2$

Take $\alpha = \pi/2$, which is not an integer multiple of π, and note $\sin\alpha = 1$. Then

$$\frac{\pi}{2} = \frac{1}{\cos\frac{\pi}{4}\cos\frac{\pi}{8}\cos\frac{\pi}{16}\cdots}.$$

It remains to show that the cosines are the nested square roots. Write $c_k = \cos(\pi/2^{k+1})$ for $k \geq 1$. Then $c_1 = \cos\frac{\pi}{4} = \sqrt{\frac{1}{2}}$. The half-angle formula $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ with $\theta = \pi/2^{k+2}$ gives

$$c_{k+1}^2 = \tfrac{1}{2} + \tfrac{1}{2}c_k, \qquad \text{so} \qquad c_{k+1} = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}c_k},$$

taking the positive root because $0 < \pi/2^{k+2} < \pi/2$, so $c_{k+1} > 0$. So

$$c_1 = \sqrt{\tfrac{1}{2}}, \quad c_2 = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2}}}, \quad c_3 = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2}}}}, \quad \ldots,$$

and each factor is the previous one fed into $t \mapsto \sqrt{\frac{1}{2} + \frac{1}{2}t}$. These are exactly the factors in the required product, so

$$\frac{\pi}{2} = \frac{1}{\sqrt{\frac{1}{2}}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}}\cdots}.$$

Checks

- With $\alpha = \pi/2$ and $n = 1$ the finite formula gives $\cos\frac{\pi}{4} = \dfrac{1}{2\sin(\pi/4)} = \dfrac{1}{\sqrt{2}}$, which is correct.
- The reciprocals of the partial products of the nested radicals are 1.4142, 1.5307, 1.5607, 1.5683, 1.5702, …; after 60 factors we get 1.5707963267949, agreeing with $\pi/2$ to machine precision.
- The finite identity also checks numerically for random α and n up to 14, and $\sin\alpha / P_{60}$ reproduces α for values such as $\alpha = 2.5$ and $\alpha = -4$.

Common slips

- Dividing by $\sin(\alpha/2^n)$ without saying why it is non-zero. This is the reason for the condition $\alpha \neq k\pi$.
- Writing $1/\lim P_n = \lim(1/P_n)$ without noting that $\lim P_n \neq 0$.
- Forgetting to justify the positive square root in the half-angle step.

Beyond STEP

The final formula is Viète's product of 1593, the first exact infinite expression for π ever written down. At university you meet infinite products properly in analysis, where a product of non-zero factors is said to converge when its partial products

tend to a *non-zero* limit, which is exactly the point we checked. Euler's product $\sin\alpha = \alpha\prod_{k\geq 1}\left(1-\frac{\alpha^2}{k^2\pi^2}\right)$ is a deeper relative. Abbott's *Understanding Analysis* is a friendly place to see these limits handled rigorously.

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